SpecialistRoute Handover — 6. Results and Findings

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On the official 236-image PlantDoc test set, the assembled pipeline reaches **72.80% ± 1.57%** as a single model (5 seeds) and **75.42%** as a 5-seed ensemble. That is comparable to, not better than, the strongest rigorously evaluated published result (76.97% ± 1.48%).

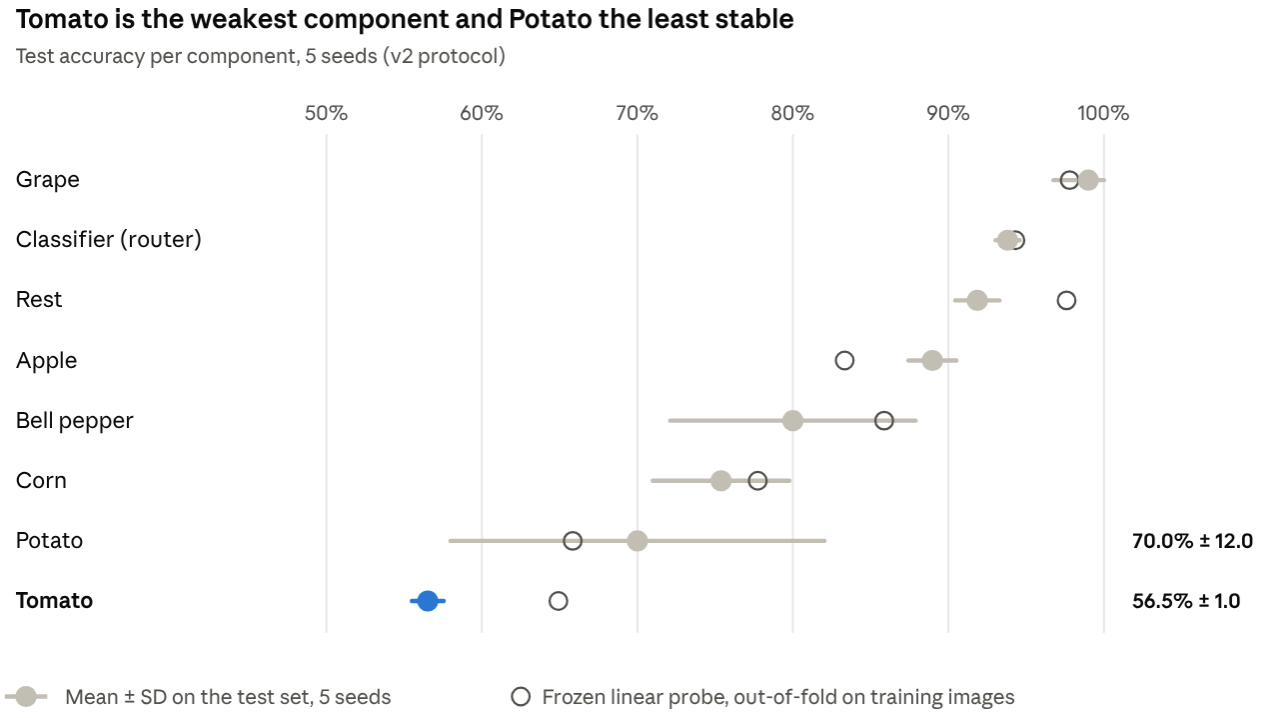
Headline results

Official test set, 236 images. Single model = mean ± SD over seeds 42, 123, 2024, 779 and 1027. Ensemble = softmax average of all 5 seeds, hard routing, no TTA.

Measure	Single model (5 seeds)	5-seed ensemble
End-to-end accuracy, all 236	**72.80% ± 1.57%** (95% CI 70.85–74.74)	**75.42%** (178/236; bootstrap CI 69.92–80.51)
Routing accuracy (species)	94.63% ± 0.84%	94.49%
Accuracy on the 216 original test images	76.02% ± 1.80%	78.70%
Accuracy on the 20 restored test images	38.0% ± 5.7%	40.0%
Seeds individually (all 236)	70.76% to 74.58%	—

The two CIs measure different things. The single-model CI is variation between seeds with the test set fixed. The ensemble's bootstrap CI is uncertainty from having only 236 test images; a single seed's bootstrap CI would be about as wide (± 5.5 pp). Report both, labelled, and state the ensemble's 5× inference cost.

Per-component accuracy



v2 component progress.json (5 seeds) and label-quality summary · test set 236 images

Each component is scored on its own test images, assuming perfect routing. Grape's spread is cut at 100%. Potato's ± 12 pp spread comes from only 16 test images, 5 of them restored. The hollow rings show that a frozen linear probe ranks the components almost the same way (next sections).

Where the errors are

Tomato causes about half of all errors, so it is where accuracy can still be gained. Per species, 5-seed ensemble, all 236 test images:

Species	Test images	Routed correctly	Correct end to end	End-to-end accuracy
Tomato	69	66	39	56.5%
Rest	59	55	53	89.8%
Apple	29	28	25	86.2%
Corn	26	26	21	80.8%
Grape	20	20	20	100%
Bell pepper	17	16	12	70.6%

Species	Test images	Routed correctly	Correct end to end	End-to-end accuracy
Potato	16	12	8	50.0%

- **Specialists cost far more than routing.** For seed 42, perfect routing would give 79.24%: routing costs about 4.7 pp, the specialists about 21 pp. Of seed 42's 60 errors, 29 are Tomato specialist errors and 13 are routing errors.
- **Tomato's weak classes:** mosaic virus and bacterial spot, confused with mold, healthy leaf and Septoria. Late blight and Septoria are recognised well.
- **Potato is a routing problem as well:** 4 of 16 Potato test images go to Tomato or Rest. Potato and Tomato share diseases (early and late blight) and have similar leaves.
- **Rest** loses most of its errors to routing (3 images sent to Apple for seed 42).
- **Hard core:** 31 test images were wrong for every seed (16 of them Tomato; earlier run, before the Potato fix).

v1 to v2: what inflated the old numbers

The v1 pipeline reported 84.27%. About 72% of the drop to v2 is evaluation methodology, and about 28% is the 20 hard test images.

Step	Accuracy (single model, 5 seeds)	Change	What changed
v1	84.27%		Checkpoints chosen on test, train–test near-duplicates present, 216 test images
v2 method, same 216 images	76.02% ± 1.80%	**−8.25 pp**	Validation-based selection, leakage removed, smaller training set
v2 method, all 236 (official)	72.80% ± 1.57%	**−3.22 pp**	The 20 images that fail the quality criteria are back

- **The 20 restored images are hard because of image quality, not their species.** Given their species mix, the ensemble would be expected to get about 14 of 20 right (≈ 70%); it gets 8 (40%). Every seed scores 30–45% on them.
- **They cluster in Potato and Tomato:** 5 of Potato's 16 test images and 7 of Tomato's 69 are restored. This partly explains Potato's ± 12 pp seed spread.

- **The 8.25 pp methodology part is not split further.** It mixes test-set selection, leakage and the smaller training set. One extra run that logs test accuracy every epoch (never used for selection) would separate the selection part.
- **Validation is easier than test** for most components (ensemble: 80.86% on validation vs 75.42% on test). The test set was never cleaned, so it is a slightly different distribution.

Ensemble and TTA choice

The variant was chosen on validation (209 images) before test was scored. The 5-seed ensemble adds about 3 points; flip TTA and soft routing add nothing that counts.

TTA	Routing	Single-seed mean (val)	5-seed ensemble (val)	Ensemble routing (val)
No	Hard	77.99%	80.86% (169)	93.30%
No	Soft	78.09%	81.34% (170)	93.30%
Flip	Hard	78.09%	80.86% (169)	92.34%
Flip	Soft	77.80%	81.34% (170)	92.34%

- **Chosen: no TTA, hard routing.** Soft routing won by exactly 1 image, and the rule fixed in advance keeps the simpler option on a gain of 1 image or less.
- **Accuracy by number of seeds averaged** (mean over all subsets): validation 77.99 → 79.38 → 80.05 → 80.48 → 80.86%; test 72.80 → 74.92 → 75.34 → 75.93 → 75.42%. The test dip from 4 to 5 seeds is noise on 236 images. Never pick a seed subset by test.
- **The ensemble beats every single seed,** on validation (169 vs a best of 167) and on test (178 vs a best of 176).

Label quality and data difficulty

PlantDoc's labels look largely right; what limits accuracy is how hard the images are, and that difficulty does not depend on the model. Each training image was scored by a model that never saw it: frozen ImageNet ConvNeXt-Large features, 5-fold logistic regression, then cleanlab.

Component	Probe accuracy (out-of-fold)	Flagged training images
Grape	97.8%	0 (0%)
Rest	97.6%	0 (0%)
Classifier (species)	94.3%	37 (2.0%)

Component	Probe accuracy (out-of-fold)	Flagged training images
Bell pepper	85.9%	6 (6.5%)
Apple	83.3%	17 (8.3%)
Corn	77.7%	41 (14.0%)
Potato	65.8%	41 (25.5%)
Tomato	64.9%	134 (26.0%)

- **Model-independent difficulty:** a frozen linear probe ranks the components almost exactly as the fully trained specialists do (only Apple and Bell pepper swap; see the hollow rings in the per-component chart).
- **Species labels are clean:** 2.0% flagged overall. Potato is the exception, with 21 of 161 images (13%) flagged at species level; this fits its Potato → Tomato routing errors. Those 21 images have not been reviewed.
- **Disease-level flags track difficulty:** the flag rate rises with probe error (roughly 0.75 × error), so they mostly mark hard images, not wrong labels. Weakest classes: Tomato early blight (29% probe accuracy) and bacterial spot (43%), Corn gray leaf spot (48%).
- **Cleaning didn't help the probe:** the same probe was trained on all labels, quality-weighted labels, and with flagged images removed. Neither cleaned version beat all labels on validation log-loss; removing flags hurt Apple (0.18 → 0.50). For Tomato, Potato and Corn this is inconclusive, because a frozen probe is a weak judge of fine-grained diseases.
- **Not yet done:** the human review of flagged images (`<Component>_review.csv`, answer y = wrong label, n = right, u = can't tell), and checking that the folds really kept look-alike pairs together (every component reported 0 pairs grouped).

Key findings for the paper

1. **Evaluation shortcuts inflated PlantDoc accuracy by 11.5 pp** in this project (84.27% → 72.80%): 8.25 pp from methodology, 3.22 pp from 20 low-quality test images. This is the strongest methodological contribution.
2. **A 7-class router with an explicit Rest class beats a 6-class router with a confidence threshold** (+18.5 pp in v1).
3. **Confidence-based re-routing between specialists does not work,** because specialists are overconfident on images from other species (negative result).
4. **Seed ensembling gives +2.6 pp on test** (72.80% → 75.42%) and beats every single seed; flip TTA and soft routing add nothing.

5. **Pipeline accuracy can be predicted from the components.** Router recall × specialist accuracy, weighted by test-set counts, matches the measured value within 0.1–0.4 pp.
6. **Difficulty is in the data, not the model:** a frozen linear probe reproduces the trained specialists' difficulty ranking, and species labels are clean (2.0% flagged).
7. **Images that fail the quality criteria are about twice as likely to be wrong** (38% vs 76% accuracy), beyond what their species predicts.
8. **Tomato is the bottleneck** (about half of all errors; 56.5% end to end), and **Potato is confused with Tomato at the routing stage.**

State with every result: the protocol (official test set, held-out validation, val_loss selection, 5 seeds), that presets came from v1, the Potato backbone change, and the fine-tuning caveat (Known Issues).

Literature context

Only results on the official split with held-out validation are directly comparable; against those, SpecialistRoute is comparable, not state of the art.

Work	Reported accuracy	Protocol	Comparable?
El Karch et al. 2026, Frontiers in Plant Science	76.97% ± 1.48%	Official split, held-out validation, 5 seeds	**Yes**: the reference to cite
Wei et al. 2024 (PlantWild), CLIP ViT-L/14	77.23%	Their own 70/10/20 split	Partly: different split
Salman et al. 2025, ViT + Mixture of Experts	74%	80/10/10 split	Partly
Singh et al. 2020, original PlantDoc paper	70.53%	Cropped PlantDoc, PlantVillage transfer	No: different task variant
PND-Net	84.30%	Split counts don't match the official one	No
ConvGeM-NeXt	94.69%	Random 70/15/15 re-split	No: likely leakage-prone

- **Our comparable numbers:** 72.80% ± 1.57% (single model) and 75.42% (ensemble, 5× inference). El Karch's 76.97% sits inside our CI, so it is a statistical tie.
- **Novelty risk:** species-then-disease chaining has been published on PlantDoc (Mo-DsCC / GSMo-CNN). Frame the paper as **methods-led**: the protocol, the inflation breakdown, the router design and the data-difficulty evidence, not "first hierarchical

model".

- The v1 literature comparison document claimed state of the art with 84.27%. That claim no longer holds; rewrite it with the numbers above.